

Jushen Dhillon

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[Github](#) | [LinkedIn](#) | [Traject](#) | [Portfolio](#)

EDUCATION

University of Southern California

Expected Graduation: August 2027

B.S. in Computer Science and Business Administration; AI Emphasis Certificate expected

Relevant Coursework: Data Structures & Object-Oriented Design in C++, Algorithms, Software Development in Java, Software Engineering, Natural Language Processing, Applied Neural Networks

Planned Coursework — Spring 2027: Operating Systems; CSCI 401 Capstone: Design & Construction of Large Software Systems; Essentials & Digital Frontiers of Big Data

WORK EXPERIENCE

BusinessWise — AI / Software Engineering Consultant

July 2026 – Present

- Building a company-intelligence pipeline ingesting multi-source Atlanta business data, performing entity resolution against a 399K+ record legacy database, and routing new, duplicate, and ambiguous entities through enrichment and human review.
- Engineered asynchronous enrichment infrastructure with durable job processing, idempotent workflows, bounded retries, exponential backoff, crash recovery, and provenance-aware validation across company, contact, and location data.
- Developed ingestion and normalization workflows across county occupational-tax, NPPES, and WARN data, integrating backend classification APIs with a human-in-the-loop review interface for evidence, conflicts, missing fields, and publication readiness.

Traject — Co-Founder & Founding Engineer

July 2025 - Present

- Built and deployed Traject end-to-end using Next.js, TypeScript, PostgreSQL/Supabase, AWS Lambda, and AI retrieval pipelines, supporting transcript-aware academic planning, course sequencing, prerequisite validation, and transfer pathway generation.
- Designed backend and data logic to normalize inconsistent articulation and degree-requirement data across institutions and translate it into personalized semester-by-semester course plans.
- Developed an AI academic advisor using OpenAI, embeddings, and custom retrieval/context pipelines to answer questions about course requirements, academic risk, plan constraints, and transfer readiness.
- Grew the platform to approximately 35 early users and conducted product demonstrations with students, advisors, success coaches, and Pasadena City College stakeholders, incorporating institutional feedback into product and system design.

Akala — Frontend Engineering Intern

June - August 2025

- Rebuilt core application flows and responsive interfaces using React, TypeScript, TanStack Router, and Shadcn, implementing frontend functionality against existing backend API contracts and product requirements.
- Worked in a 3-person Agile engineering team using GitHub pull requests, code reviews, sprint planning, and iterative feature delivery.

Acumen Velocity — Software Engineering Intern

June - August 2024

- Built Java/Spring Boot healthcare-data workflows to transform CSV/JSON records into HL7 FHIR-compliant resources and support patient-data search, editing, and retrieval.
- Integrated Google Cloud Healthcare API and Google OAuth authentication to support secure cloud-based healthcare-data access and application workflows.

SKILLS

Languages: Java, Python, C++, TypeScript, JavaScript, SQL

Backend / Data: Node.js, Spring Boot, PostgreSQL, Supabase, MongoDB, REST APIs

Cloud / Infrastructure: AWS Lambda, Google Cloud, Docker, Railway

AI / ML: OpenAI API, Embeddings, RAG Pipelines, ChromaDB

Frontend: React, Next.js, Tailwind CSS

Testing / Tools: Git, Jest, Postman